

5

Simulated antenna input and runtime behaviour

DONE | TOTAL: orig ~6d · total ~0d | completed 09 Oct 2026

GOAL

Process repeatable simulated antenna input through one TimingNode using normal runtime, domain and persistence paths.

RESULT

- Simulated antenna observations reach the normal registration path.
- Successful TimingNode OPEN/CLOSE transitions are represented in the normal committed
- Registration revoke appends REV through the same committed source stream and leaves the
- Normal manual-registration ADD is available through IF-03 with client-supplied
- Simulated-tag profiles drive the normal antenna path rather than direct registration.
- Sustained input can be measured without bypassing TimingNode ownership.
- Restart/recovery works with the same simulated input path used by automated tests.

END DEMO

- Open one TimingNode and show the committed OPEN TimingData record.
- Feed repeatable tag observations through SimulatedAntenna and show which observations
- Add manual registrations using both client-selected current time and operator-entered
- Run a short simulated-tag batch and show one representative two-tag passage reaching
- Revoke one registration through IF-03 and show durable ADD plus REV source
- Close the TimingNode and show the CLOSE record in the same source sequence.
- Inspect the runtime counters, restart SI-01 and continue using the same simulated input

DOCUMENTATION

- 30-UC R-80
- 41-01-SSD R-70
- 32-03-IDD R-80
- 60-SVP R-70

TOOLING / ENGINEERING ENVIRONMENT

T01 after D01

DONE

Runtime-characterization harness and evidence tooling

Completed through Java PR...

DOCUMENTATION / DECISIONS

D01

DONE

Runtime execution and measurement plan

Define measurement questions,...

D02

DONE

Define OPEN/CLOSE TimingData semantics and reference mapping

Defines successful CLOSED-OPEN...

D03

DONE

Define registration revoke semantics and public contract

Defines append-only...

D04 after D01

DONE

Review simulated antenna/input architecture

Review the simulated...

D05 after D01

DONE

Define internal runtime-observability architecture

Define the internal...

D06

DONE

Link SSD architecture Needs to detailed SDD design

Completed through meta issue...

D07

DONE

Clarify Presentation external ports in architecture diagram

Completed through meta issue...

D08

DONE

Define manual-registration API and timestamp precision

Define normal IF-03...

D09 after D04

DONE

Define simulation layers, simulated tags and profile control

Distinguish direct accepted-re...

APPLICATION / PRODUCT

A01 after D04

DONE

Qualify/implement simulated antenna and tag-processing path

Requalified the simulated-inpu...

A02 after D05 A01

DONE

Qualify/implement runtime markers and counters

Requalified and corrected on...

A03 after V01

DONE

Measurement-driven allocation/data-access decision

V01 development-host evidence...

A04 after D02

DONE

Commit OPEN/CLOSE through the normal TimingData path

Commit successful TimingNode...

A05 after D03

DONE

Commit registration revoke through the normal TimingData path

Add the public node-scoped...

A06 after D08

DONE

Implement manual-registration API, precision policy and Development Client controls

Completed through Java issue...

A07 after D09

DONE

Implement simulated-tag profiles and Development Client batch simulation

Completed through Java issue...

VERIFICATION / TEST

V01 after D01 D04 D05 T01 A01 A02

DONE

Single-node baseline load/burst characterization

Development-host baseline is...

V02 after A03

DONE

Sustained tag-observation load and stalled-downstream fairness/backpressure

Completed through Java issue...

V03 after A01

DONE

Restart and recovery with simulated input

Completed through Java PR...

V04 after A01

DONE

Provider bootstrap verification

Verified built-in EventDataPro...

V05 after A04

DONE

OPEN/CLOSE TimingData ordering, persistence and rejection verification

Completed through Java PR...

V06 after A05

DONE

Registration revoke/API/persistence verification (client UI evidence in Step 6 V01)

SI-01 revoke/API/persistence...

V07 after A06

DONE

Manual registration, timestamp precision and Development Client projection verification

Completed with the automated...

V08 after A07

DONE

Simulated-tag profile and batch-driver verification

Completed through Java issue...

PLANNING CHANGES

- 2026-10-09 — Complete Step 5 on the proven SI-01/server verification baseline. V06 closes on revoke/API/persistence evidence; GUI revoke/DELETED/reconnect/recovery verification moves to Step 6 V01 using SI-02.
- 2026-10-07 — Add D08/A06/V07 for normal manual-registration semantics and time precision, and D09/A07/V08 for deterministic simulated-tag profiles and batch control.
- 2026-10-07 — Move selected-target runtime evidence to Step 9. Step-5 V01 is the development-host baseline and releases the A03 runtime design decision.
- 2026-10-07 — Complete T01 through Java PR #327 and make V01 runtime baseline characterization the next activity.

PLANNING CHANGES (CONT.)

- 2026-10-06 — Requalify A01 on Java main and complete V04 provider bootstrap verification through PR #302 with built-in reference providers and an external provider JAR.
- 2026-10-04 — Complete D05 runtime-measurement design while keeping A01 active until the antenna/tag-processing path is wired into runtime composition.
- 2026-10-04 — Reopen D01 and decompose the work into D04/D05 design gates plus T01 harness; A01/A02 require requalification and V01 moves before A03.
- 2026-10-04 — Fix the single-TimingNode runtime-measurement baseline in D01 and make A01 the active implementation activity.

PLANNING CHANGES (CONT.)

- 2026-10-04 — Activate Step 5 after Step-4 VC-ST1-003 closure with D01 as the first planning/design activity.